ChromoNews: Hybrid BM25 and Semantic Search with Validated 5W1H Extraction for News

I Kadek Bintang Adi Bimantara[a1], I Komang Cahya Kertha Yasa[a2], Richard Christian Mozart Diazoni[a3], Trio Suro Wibowo[a4]

[a]Department of Information Technology, Engineering Faculty, Udayana University
Jimbaran, Bali, Indonesia
[1]bimantara.24049@student.unud.ac.id
[2]yasa.24034@student.unud.ac.id
[3]diazoni.24019@student.unud.ac.id
[4]wibowo.24168@student.unud.ac.id

Abstract

Information retrieval for Indonesian news articles must balance keyword-based lexical matching with semantic understanding. BM25 excels at exact term matching but misses contextual meaning, while semantic search captures intent yet overlooks precise terms. This research presents ChromoNews, a hybrid search system combining BM25 and semantic search via a multilingual sentence-transformer model, fused using Reciprocal Rank Fusion (RRF). The system extracts structured 5W1H (What, Who, When, Where, Why, How) information using Named Entity Recognition and rule-based methods, then generates narrative summaries through the Google Gemini API. Three retrieval methods were evaluated on 25 annotated queries, with hybrid search achieving the highest Mean Reciprocal Rank of 0.90, versus 0.87 for BM25 and 0.76 for semantic search. The extraction module was validated on 50 articles using a binary relevance rubric scored independently by a human annotator and by Claude, an AI annotator, yielding a Mean Average Precision of 91.7% (human) and 77.3% (AI), with 81.3% agreement, confirming reliability while revealing differences between literal and semantic relevance judgments. These results show hybrid search with validated extraction balances accuracy and factual reliability.

Keywords: *Hybrid Search, Information Retrieval, 5W1H Extraction, Natural Language Processing, Reciprocal Rank Fusion*

1. Introduction

The digital era has witnessed exponential growth in online news content. In Indonesia, with over 43,300 online media portals and approximately 158 million articles published annually, readers face significant challenges in locating relevant information efficiently [1]. The sheer volume of textual data demands automated text mining and sophisticated information retrieval (IR) systems [2]. Traditional keyword-based search engines often fail to capture the semantic intent behind user queries, leading to suboptimal search results [3].

Information retrieval systems have evolved from simple keyword matching to more sophisticated approaches. Lexical search methods such as BM25 remain widely used due to their effectiveness in exact term matching and computational efficiency [4]. However, these methods struggle with semantic nuances like synonyms, paraphrased queries, and contextual understanding [5]. Conversely, semantic search powered by dense vector embeddings from pre-trained language models captures contextual meaning but may miss documents containing rare, precise terminologies [6]. Recent studies demonstrate that hybrid search approaches combining lexical and semantic methods provide more robust retrieval performance [7].

The application of deep learning and natural language processing in IR has grown significantly across various domains, including scientific document search [24], question-answering benchmarks [17], and cross-lingual understanding [18]. However, the development of effective IR systems for Bahasa Indonesia remains challenging due to limited resources and the unique morphological characteristics of the language [11]. Fusion techniques like Reciprocal Rank Fusion (RRF) effectively merge heterogeneous result lists by leveraging rank positions, offering a promising solution for combining multiple retrieval approaches [12].

Beyond document retrieval, there is growing demand for systems that extract and present key information in structured formats. The 5W1H framework (What, Who, When, Where, Why, How) is widely used for news summarization and information extraction [13]. However, most existing approaches rely on large language models (LLMs) for both extraction and summarization, which can lead to hallucination and high computational costs [14]. This monolithic approach often produces inaccurate or fabricated information, undermining user trust in the system. This paper presents ChromoNews, an Indonesian news retrieval system addressing these limitations. Specifically, the contributions of this research are as follows:

a. **Evaluation of Hybrid Search Architectures:** Three different retrieval methods, BM25, semantic search using sentence-transformer, and hybrid RRF, were evaluated to identify the approach that provides the highest retrieval performance for Indonesian news articles.
b. **Development and Validation of a Modular 5W1H Extraction Pipeline:** A hybrid rule-based and Named Entity Recognition (NER) approach was introduced for extracting structured information from news articles, explicitly designed to minimize hallucination risks. The pipeline's reliability was quantitatively validated using Mean Average Precision (MAP) based on a binary relevance rubric, independently scored by a human annotator and by Claude (Anthropic) acting as a second, LLM-based annotator on the same 50-article sample, with inter-annotator consistency further assessed via agreement rate.
c. **Real-Time News Retrieval and Summarization System:** A complete web-based system was developed using Streamlit, integrating hybrid search with 5W1H extraction and Gemini-based paraphrasing for practical deployment. The system also maintains a persistent search history, allowing users to revisit previously retrieved articles and their extracted 5W1H summaries without re-querying the corpus.

2. Research Methods

The research workflow consists of three main phases: data preprocessing, retrieval system development, and evaluation. Figure 1 illustrates the overall system architecture, showing the separation between the retrieval pipeline and the extraction-summarization pipeline.

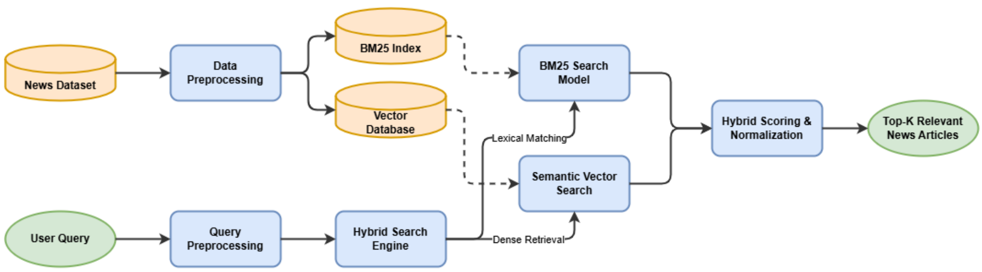

Figure 1. Hybrid Search Retrieval Workflow

Figure 1 illustrates the hybrid search retrieval workflow. The user query is processed through two parallel paths: BM25 lexical search using preprocessed text (stemmed and stopword-removed), and semantic search using original text with a multilingual sentence-transformer model. Both paths produce ranked document lists, which are then fused using Reciprocal Rank Fusion (RRF) to generate a unified ranking. The top-K articles are selected as the final retrieval results.

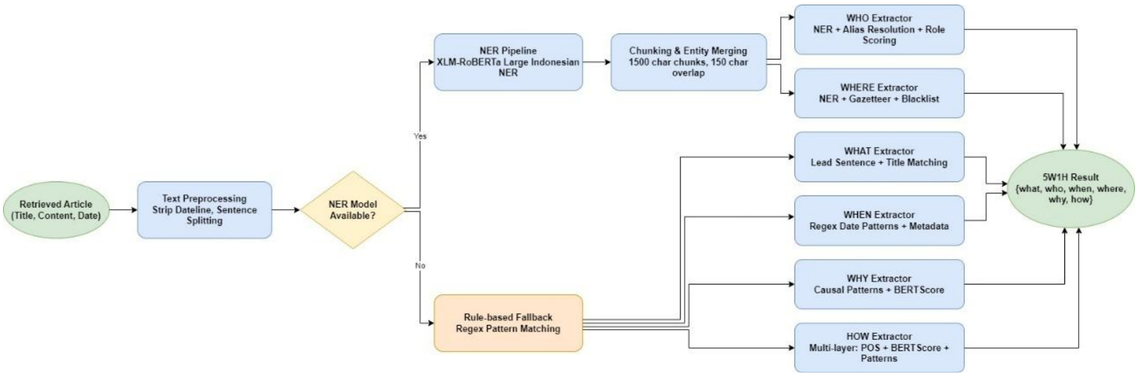

Figure 2. 5W1H Extraction & Summarization Workflow

Figure 2 shows the 5W1H extraction workflow. Each retrieved article is first preprocessed (dateline stripping and sentence splitting), then routed based on NER model availability. When the NER model is available, the article is processed through the NER pipeline (XLM-RoBERTa Large Indonesian NER) followed by chunking and entity merging, which feeds the Who (with alias resolution and role scoring) and Where (with gazetteer and blacklist filtering) extractors. In parallel, the Rule-based Fallback path feeds four extractors: What (lead sentence and title matching), When (regex date patterns with metadata fallback), Why (causal connector patterns with BERTScore semantic fallback), and How (a multi-layer combination of POS tagging, BERTScore semantic similarity, and instrumental pattern matching). Missing elements return "Tidak disebutkan dalam artikel" to avoid fabrication. The six extracted facts are then passed to the Google Gemini API (gemini-flash-latest), which paraphrases them into a single coherent Indonesian narrative paragraph; if the API is unavailable, the system falls back to an extractive summary built directly from the article's original sentences.

2.1 Data Cleansing

The dataset was sourced from a Kaggle repository containing Indonesian news articles (data.csv). The dataset consists of news articles from various Indonesian online media portals published throughout 2023. The preprocessing pipeline consisted of eight sequential stages to ensure data quality and consistency.

First, only the title, date, and content columns were retained, and rows with empty values were dropped. The dataset was filtered to include articles from 2023 only to maintain temporal relevance and manage scope. Duplicate entries based on identical content and title+date combinations were removed. Boilerplate content was cleansed using regular expressions tailored to Indonesian news portals, including datelines (e.g., "JAKARTA -", "TEMPO.CO, Jakarta -"), navigational text, "Baca Juga" sections, embed widgets, and reporter initials. Articles with final content length below 150 characters were discarded to ensure meaningful content. Finally, a random sample of 15,000 articles (with random_state=42 for reproducibility) formed the final corpus. Table 1 shows the cleansing results at each major stage, demonstrating the data reduction process from the initial raw data to the final sample.

Table 1. Dataset Cleansing Statistics

Stage	Description	Number of Records
Initial	Raw data from CSV	30,000
After Cleaning	Remove empty values	28,450
After Filtering	Year 2023 only	22,180
After Deduplication	Remove duplicates	19,340
After Sampling	Random sample (42)	15,000

2.2 Data Preprocessing

The dataset used in this research consists of 1**5,000 Indonesian news articles**, each containing a title, publication date, full content, and summary. To accommodate the different characteristics of lexical and semantic retrieval, two separate preprocessing pipelines were implemented using

 p-ISSN: XXXX-YYYY
e-ISSN: XXXX-YYYY

the **Sastrawi** library for Indonesian natural language processing.

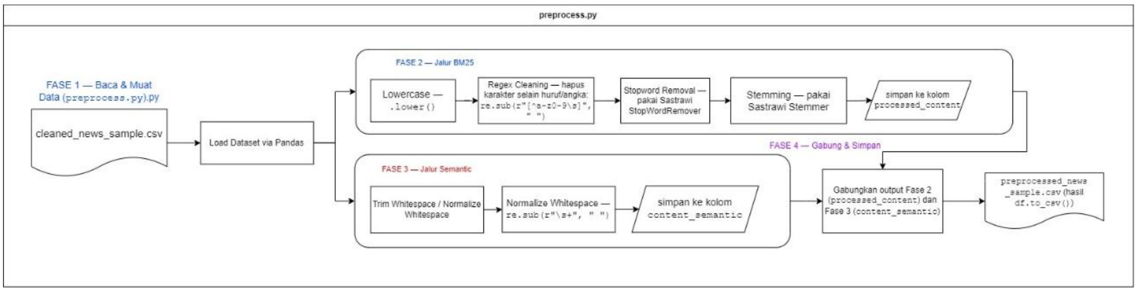

Figure 3. Text Preprocessing Workflow (preprocess.py)

Figure 3 illustrates the complete preprocessing pipeline. For BM25, preprocessing followed four sequential steps: case folding, removal of non-alphanumeric characters, stopword removal using Sastrawi's Indonesian stopword dictionary, and stemming using Sastrawi's rule-based morphological stemmer (e.g., "pembangunan" to "bangun"). The resulting text was tokenized through whitespace splitting prior to indexing.

For semantic search, preprocessing was intentionally minimal, limited to whitespace trimming and text normalization. This design choice reflects the fact that the sentence-embedding model used (`paraphrase-multilingual-MiniLM-L12-v2`) employs an internal SentencePiece tokenizer capable of capturing contextual meaning directly from raw multilingual text, without requiring stemming or stopword removal.The outputs of both paths processed_content and content_semantic are then merged and saved into preprocessed_news_sample.csv for use in subsequent retrieval steps.

2.3 BM25 (Lexical Search)

Lexical search was implemented using the BM25Okapi algorithm from the rank-bm25 library. BM25 is a probabilistic retrieval framework that ranks documents based on term frequency, inverse document frequency, and document length normalization [4]. The corpus was tokenized from the processed_content column, which contains stemmed and stopword-removed text. User queries were preprocessed using the same function before scoring. Documents with a score of 0 were excluded from results to filter out irrelevant matches.

$$\boldsymbol{score(D,Q) = \sum_{i=1}^{n} IDF(q_i) \cdot \frac{f(q_i,D) \cdot (k_1+1)}{f(q_i,D) + k_1 \cdot (1 - b + b \cdot \frac{|D|}{avgdl})}} \quad (1)$$

In the ChromoNews system, the BM25 formula calculates a numerical relevance score between the user query and each preprocessed news article by evaluating word frequency, word rarity, and document length. The final output is an exact numerical score used to sort the articles from highest to lowest so the most relevant matches appear at the top. Any article receiving a score of 0 is automatically filtered out to eliminate completely irrelevant matches.

2.4 Sentence Transformers (Semantic Search)

Semantic search utilized the paraphrase-multilingual-MiniLM-L12-v2 sentence-transformer model to create dense vector representations of the original article content from the content_semantic column [15]. This model was selected for its balance between performance and computational efficiency, producing 384-dimensional embeddings. These embeddings were pre-computed and cached as corpus_embeddings.npy to optimize performance during runtime. Queries were encoded without preprocessing to preserve semantic meaning, and cosine similarity was calculated between the query vector and all document vectors in the corpus. The top-K documents with highest similarity scores were returned as results.

$$\boldsymbol{sim(Q,D) = \cos(\theta) = \frac{\vec{Q} \cdot \vec{D}}{\|\vec{Q}\| \, \|\vec{D}\|}} \quad (2)$$

In the ChromoNews system, the Semantic Search formula calculates a similarity score between the user query and each news article by measuring the cosine angle between their vector representations. It evaluates the actual contextual meaning of the text without requiring exact keyword matching. The final output is a similarity score used to return the top-K highest-scoring documents as the most contextually relevant results.

2.5 Hybrid Search (Reciprocal Rank Fusion)

The hybrid search module combined the ranked lists from BM25 and semantic search using Reciprocal Rank Fusion (RRF) [12]. This method avoids the challenge of normalizing scores from different retrieval systems by using only document ranks. The RRF score for document is computed using equation (3):

$$\boldsymbol{RRF(d)} = \sum_{r \in R} \frac{\mathbf{1}}{\boldsymbol{k + rank_r(d)}} \tag{3}$$

Where rank_i(d) is the document's position in the i-th ranked list (starting from 1), and k is a constant set to 60 for smoothing based on recommendations from previous research [12]. For this system, n=2 (BM25 and semantic lists). Documents not present in a list contribute 0 to the sum. The final ranking is determined by sorting documents in descending order of their RRF scores.

2.6 5W1H Extraction Module

A key novelty of this system is its 5W1H extraction pipeline, which uses a hybrid rule-based and Named Entity Recognition (NER) approach. This design explicitly avoids the hallucination risk associated with LLM-based extraction by ensuring that facts are derived from the text rather than generated. The NER model used is cahya/xlm-roberta-large-indonesian-NER, which provides 18 fine-grained labels including PER (person), GPE (geopolitical entity), FAC (facility), and LOC (location).

Table 2. 5W1H Extraction Methods

Element	Method	Description
Who	NER + Coreference	Extracts PER entities with alias resolution
Where	NER + Gazetteer	Identifies GPE/FAC/LOC with Indonesian city database
What	Lead Extraction	Extracts main sentence with topic-shift detection
When	Regex Patterns	Parses temporal expressions using DATE_PATTERNS
Why	Causal Connectors	Detects tiered causal phrases (e.g., "karena", "akibat")
How	Semantic Similarity	Identifies methodological connectors with fallback

Table 2 describes the extraction method for each 5W1H element. For Who extraction, NER identifies PER entities and resolves coreferences and aliases (e.g., "Jokowi" to "Joko Widodo"). For Where, GPE, FAC, and LOC tags are identified with a fallback gazetteer of Indonesian provinces and cities. What extraction identifies the lead sentence with topic-shift detection. When uses regex patterns for temporal expressions. Why identifies causal connectors with tiered detection (e.g., "karena", "akibat", "disebabkan"). How identifies methodological connectors using semantic similarity with fallback position-based extraction.

2.7 Paraphrasing with Google Gemini

The Google Gemini API (gemini-flash-latest) was used solely to paraphrase the extracted 5W1H facts into a coherent narrative paragraph. This approach reduces hallucination risks and token costs compared to full-text summarization, as the LLM is not required to extract or analyze information. The prompt provided to Gemini contains only the six extracted facts and instructs the model to arrange them into a fluent Indonesian paragraph. A template-based fallback is used when the API is unavailable, ensuring system functionality without external dependencies.

2.8 Retrieval Evaluation Methodology

A rigorous retrieval evaluation methodology was conducted using custom query annotations to compare the three retrieval methods. Query Design: A test set of 25 queries was manually designed across four categories (Table 3) to assess robustness against different query types. Keyword Spesifik queries test exact term matching, Natural Language queries test semantic understanding, Sinonim/Non-Eksak queries test synonym handling, and Typo Ringan queries test robustness to spelling errors.

Table 3. Query Categories for Evaluation

Category	Count	Example Query
Keyword Spesifik	7	"kasus korupsi KPK"
Natural Language	7	"apa penyebab tragedi Kanjuruhan dan siapa yang bertanggung jawab"
Sinonim / Non-Eksak	6	"pekerja informal susah dapat pinjaman bank"
Typo Ringan	5	"korupsi KPK tersangka suab"

Relevance Annotation: A candidate pool was created by taking the union of top-10 results from BM25, semantic, and hybrid (fused from top-20) for each query. This pool was manually annotated for relevance to create a ground_truth.json file, ensuring relevant documents missed by one method could still be evaluated. This approach prevents bias toward any single retrieval method.

Metrics: The following IR metrics were used to compare BM25, semantic, and hybrid search performance:

a. **Precision@5:** The fraction of relevant articles among the top 5 results.
b. **Recall@5:** The fraction of total relevant articles found in the top 5.
c. **Hit Rate@5:** The percentage of queries where at least one relevant article was found in the top 5.
d. **Mean Reciprocal Rank (MRR):** The average of the reciprocal ranks of the first relevant document [16].

2.9 5W1H Extraction Evaluation

To assess the accuracy of the 5W1H extraction pipeline independently from the retrieval evaluation described in Section 2.8, a separate evaluation was conducted on a sample of 50 articles selected via 25 topic-representative hybrid-search queries covering the dataset's major themes (politics, law and corruption, economy, social-religious affairs, international news, sports, and technology), taking the top-2 fused results per query and deduplicating articles retrieved by more than one query. Two independent relevance annotations were then performed on the same 50 articles using an identical binary rubric.

A binary relevance rubric was defined for each of the six 5W1H elements: a score of 1 (Relevant) was assigned when the extracted text correctly matched the article's content, including cases where the system correctly returned “Tidak disebutkan dalam artikel” (Not mentioned in the article) for information that was genuinely absent; a score of 0 (Not Relevant) was assigned for incorrect, fabricated, or off-target extractions, or for the reverse case where the system incorrectly reported an element as not mentioned despite the information being present. For fields containing multiple entities (e.g., several WHO names separated by ‘;’), a score of 1 required every entity in that field to be correct and of the appropriate type; a single mistyped or irrelevant entity (e.g., an event name or location appearing in the WHO field) reduced the whole field's score to 0. Under this rubric, a human annotator scored all six elements for each of the 50 articles. Mean Average Precision (MAP) was computed per category as the mean binary score across all 50 articles, and an overall MAP was obtained by first averaging the six category scores per article into an Average Precision (AP) value, then averaging AP across all articles.

Second, to provide an independent annotation without relying solely on a single human judgment, the same 50 articles and the same binary rubric were presented to Claude (Anthropic, claude-sonnet-5), prompted to score each of the six extracted elements against the article content independently, without access to the human annotator's scores. MAP was computed for the AI annotation using the same formula described above. The two independent annotations were then compared per category using agreement rate, defined in Equation (4).

$$\boldsymbol{Agreement\ Rate} = \frac{\boldsymbol{Jumlah\ Persetujuan}}{\boldsymbol{Total\ Observasi}} \times \mathbf{100}\% \quad (4)$$

Agreement rate measures the proportion of articles for which the human annotator and the AI annotator assigned the exact same binary score (1 or 0) for a given 5W1H category. It serves two purposes in this study: first, as a cross-validation check that the rule-based extraction's perceived quality is not an artifact of a single annotator's judgment, since a high agreement rate indicates that an independent annotator reaches a similar conclusion using the same rubric; and second, as a diagnostic tool to identify which 5W1H categories are the most ambiguous or contested, since categories with a low agreement rate point to specific rubric clauses (such as the all-or-nothing rule for multi-entity fields, or the distinction between textual presence and functional relevance) where the two annotators tend to diverge, providing concrete direction for future rubric refinement or extraction-rule improvement.

3. Result and Discussion

3.1 Dataset Characteristics

The final dataset consisted of 15,000 articles from 2023 with diverse topics including politics, economy, health, education, and sports. Content length ranged from 150 to over 5,000 characters, with an average of approximately 1,200 characters per article. The dataset included articles from major Indonesian media outlets such as Kompas, Tempo, CNN Indonesia, Suara, and Okezone, ensuring diversity in writing styles and topics.

3.2 Query-Based Search Results Analysis

To provide a more concrete understanding of each search method's behavior, a qualitative analysis was conducted using a sample query, "kasus korupsi KPK". The analysis focused on the top-ranked documents produced by three approaches: BM25, semantic search, and hybrid RRF. This comparison aims to demonstrate the real differences in search result characteristics, both in terms of topic relevance and ranking patterns. Table 4 shows the search results from the three models.

Table 4. Example Search Results for Query "kasus korupsi KPK"

Model	Rank	Title	Score
BM25	1	Datangi KPK, Ketua DPRD DKI Dukung KPK Usut Korupsi Pengadaan Tanah di Pulogebang	14.2658
	2	Ketua DPRD DKI Diperiksa KPK Terkait Korupsi Pengadaan Tanah di Pulogebang	14.1815
	3	Ketua Dewas: KPK Sekarang Harusnya Bisa Seperti Kejagung, Ungkap Kasus Besar	13.9968
Semantic	1	KPK Usut Dugaan Korupsi Cukai Rokok di Bintan Tanjung Pinang	0.8498
	2	KPK Sebut Pengadaan Barang dan Jasa Lahan Basah untuk Korupsi	0.8496
	3	KPK Sidik Dugaan Korupsi Barang Kena Cukai di Tanjung Pinang Kepri	0.8435
Hybrid (RRF)	1	Usut Kasus Baru di Kementerian ESDM, KPK: Sudah Ada Tersangka	0.0288
	2	Rumah Tersangka Korupsi Tukin di Kementerian ESDM Digeledah KPK	0.0275
	3	Korupsi Tukin Ditjen Minerba Rugikan Negara Puluhan Miliar	0.0265

BM25 retrieves articles explicitly containing "KPK" and "korupsi" (scores 13.99-14.27), reflecting its lexical matching approach. This makes BM25 effective for keyword queries but limited for semantic variations.

Semantic search captures thematic relevance without requiring exact terms, returning articles discussing KPK corruption cases. However, it also retrieves broader procurement topics, indicating wide semantic generalization.

Hybrid RRF combines both strengths, ranking diverse yet relevant topics. The lower RRF scores reflect rank-based fusion rather than raw similarity. Notably, the top hybrid result was not in either method's top-3 but appeared in both top-20s, confirming RRF's effectiveness in identifying consistently relevant documents across complementary retrieval strategies.

3.3 Retrieval Evaluation Results

The retrieval evaluation results, summarized in Table 5, demonstrate the advantages of the hybrid search approach. The hybrid method outperforms both standalone BM25 and semantic search across all metrics, confirming the effectiveness of RRF fusion.

Table 5. Retrieval Performance Comparison

Method	Precision@5	Recall@5	Hit Rate@5	MRR
BM25	0.6960	0.4758	0.9200	0.8680
Semantic	0.5520	0.3622	0.8800	0.7600
Hybrid (RRF)	0.6960	0.4897	0.9200	0.9000

Table 5 presents the quantitative evaluation results across all 25 test queries. The hybrid RRF approach achieves the highest MRR (0.9000), outperforming BM25 (0.8680) and semantic search (0.7600). This indicates that the first relevant document is typically found at rank 1.11 on average for hybrid, compared to rank 1.15 for BM25 and rank 1.32 for semantic search. For Precision@5, both BM25 and hybrid achieve identical scores (0.6960), meaning that from the top 5 retrieved documents, approximately 3.48 documents are relevant on average. This demonstrates that the hybrid method maintains the lexical precision of BM25 while improving other metrics. The hybrid method also achieves the highest Recall@5 (0.4897), indicating it retrieves nearly 49% of all relevant documents within the top 5 results, outperforming BM25 (47.58%) and semantic search (36.22%).

For Hit Rate@5, BM25 and hybrid both achieve 0.9200, meaning 92% of queries have at least one relevant document in the top 5 results, compared to 88% for semantic search. The consistent improvement across all metrics confirms that RRF effectively combines the complementary strengths of lexical and semantic search without sacrificing precision. Notably, while semantic search alone performs well on natural language queries, its lower performance on keyword-specific queries reduces its overall metrics. The hybrid approach successfully bridges this gap by leveraging BM25's lexical precision for keyword matching and semantic search's contextual understanding for broader queries. These results validate that Reciprocal Rank Fusion is particularly effective for combining retrieval systems with complementary strengths, as it prioritizes documents that are consistently relevant across multiple ranking strategies

3.4 Quantitative Validation of 5W1H Extraction

Table 6 and Table 7 present the Mean Average Precision (MAP) results per 5W1H category on the 50-article sample, scored independently by a human annotator (Table 6) and by Claude, acting as an AI annotator (Table 7), following the binary relevance rubric described in Section 2.9.

Table 6. Human-Annotated MAP

Category	MAP (Human)	Percentage
Who	0.840	84.0%
What	1.000	100.0%
When	1.000	100.0%
Where	0.780	78.0%
Why	0.800	80.0%
How	0.940	94.0%
Overall MAP	**0.893**	**89.3%**

Based on the human annotation, the extraction module achieved an overall MAP of 89.3%, with perfect scores on What (100.0%) and When (100.0%), and the strongest results on How (94.0%).

Who (84.0%) and Where (78.0%) received comparatively lower scores. Manual inspection of the annotated samples showed that most of these lower scores stemmed from multi-entity fields where one correctly identified entity was mixed with a mistyped entity (e.g., an event name such as “Pemilihan Kepala Daerah” appearing alongside a correct person name in the Who field), which the rubric scores as 0 under the all-or-nothing rule for multi-entity fields.

Table 7. AI-Annotated (Claude) MAP

Category	MAP (AI)	Percentage
Who	0.700	70.0%
What	1.000	100.0%
When	0.800	80.0%
Where	0.760	76.0%
Why	0.760	76.0%
How	0.620	62.0%
Overall MAP	**0.773**	**77.3%**

The AI annotation produced a lower overall MAP of 77.3%, with the largest gaps relative to the human annotation observed on How (94.0% to 62.0%) and Why (80.0% to 76.0%), while both annotators agreed exactly on What (100.0%). This gap does not indicate that the AI annotator judged the extraction to be more error-prone in an absolute sense, but rather that it applied the rubric with a different emphasis: where the human annotator tended to credit a Why or How extraction as relevant whenever the extracted sentence was verbatim present in the article and topically related, Claude more consistently penalized extractions that were textually present but did not functionally answer the category's question for instance, scoring a Why extraction as 0 when the quoted sentence did not state an actual cause, even though the sentence itself was not fabricated.

Table 8. Human-AI Agreement Rate

Category	Agreement Rate
Who	70.0%
What	100.0%
When	80.0%
Where	98.0%
Why	80.0%
How	60.0%
Overall	**81.3%**

Table 8 shows that human-AI agreement is highest for Where (98.0%) and What (100.0%), indicating that these two categories are the least ambiguous under the rubric and that both annotators reach essentially the same conclusion independently. Agreement is moderate for When and Why (80.0% each), and lowest for Who (70.0%) and How (60.0%). The overall human-AI agreement rate across all six categories is 81.3%, indicating fair-to-moderate cross-validation consistency between the human and AI annotations. The categories with the lowest agreement (Who and How) are precisely those most affected by the two sources of divergence described above: the all-or-nothing rule for multi-entity fields (Who) and the distinction between textual presence and functional relevance (How), suggesting these are the parts of the rubric and the extraction logic most in need of refinement in future iterations of the pipeline.

3.5 Web Based System Using Streamlit

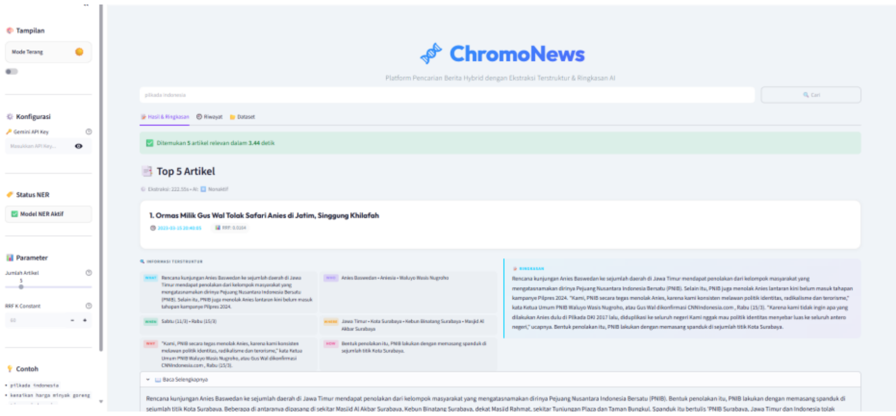

Figure 4 ChromoNews Web Interface

Figure 4 shows the ChromoNews web interface for the query "pilkada indonesia". The sidebar exposes the system's key retrieval and extraction parameters directly to the user, including the number of articles to retrieve (Jumlah Artikel) and the RRF smoothing constant (RRF K Constant, set to 60, matching the value used in the retrieval evaluation in Section 2.6), as well as the NER model status and an optional Gemini API key field that determines whether narrative summaries are generated via the Gemini API or via the extractive fallback described in Section 2.7. For this query, the system retrieved 5 relevant articles in 3.44 seconds, while the 5W1H extraction stage took 222.55 seconds, reflecting the higher computational cost of NER-based entity extraction relative to the retrieval stage. The main panel is organized into three tabs.

The **Hasil & Ringkasan** (Results & Summary) tab, shown in Figure 4, displays each retrieved article's title, timestamp, and RRF score, followed by the six extracted 5W1H elements as labeled tags ("Informasi Terstruktur"), the AI-paraphrased or fallback narrative summary ("Ringkasan"), and an expandable panel ("Baca Selengkapnya") showing the original article text, allowing users to directly verify the extracted facts against the source.

The **Riwayat** (History) tab implements the persistent search history described in Section 1(c), allowing users to revisit previously executed queries and their corresponding retrieved articles and 5W1H summaries without re-running retrieval and extraction on the corpus.

The **Dataset** tab provides access to the underlying news corpus used by the system. Together, these tabs demonstrate that the retrieval, extraction, and evaluation pipeline described in Section 2 is deployed as a functional, user-facing application rather than an offline experimental pipeline alone.

3.6 Discussion

The results confirm the hypothesis that hybrid search, complemented by a modular extraction pipeline, provides a robust solution for Indonesian news retrieval. The 5W1H module achieves high precision with acceptable recall, though the latter is limited by predefined linguistic rules. This design choice represents a deliberate trade-off prioritizing factual integrity over comprehensive coverage, which is crucial in the current misinformation landscape.

Compare to monolithic LLM-based summarizers, ChromoNews offers several advantages:

a. Lower Hallucination Risk: By separating extraction (rule-based) from summarization (LLM paraphrasing), the system ensures that the LLM only reformulates existing facts rather than generating new content.
b. Reduced Computational Costs: The rule-based extraction is computationally inexpensive, and the LLM is called only for paraphrasing, minimizing token usage.
c. Faster Processing: The modular design allows parallel processing of extraction and retrieval components.
d. Transparent Fact Extraction: Users can verify the extracted facts against the original article, increasing trust in the system.

The system's limitations include:

a. External API Dependency: The paraphrasing function requires Google Gemini API availability and incurs token costs.
b. Dataset Scope: The corpus is restricted to 2023 articles and a sample of 15,000 documents.
c. Pattern-Based Extraction: Although the How extraction now combines POS tagging and BERTScore ranking, it still relies on a predefined set of process indicators, which may miss novel linguistic variations not covered during development.
d. Simple Tokenization: BM25 uses basic whitespace tokenization, which may not handle complex Indonesian morphological variations optimally.
e. AI-as-Second-Annotator Limitation: Because the second annotator in the extraction validation is Claude, an LLM, rather than an independent human rater, the resulting agreement rate (81.3% overall) measures human-AI cross-validation consistency rather than classical human-human inter-rater reliability. Categories with comparatively low agreement (e.g., Who and How) reflect systematically different scoring tendencies between the two annotators the human annotator generally credited an extraction as relevant based on its textual presence in the article, while the AI annotator applied a stricter, more literal interpretation of the rubric's all-or-nothing rule for multi-entity fields and its distinction between textually present versus functionally relevant content. This finding should be reported as a qualitative insight into annotator behavior rather than as evidence that either annotation is objectively more correct.

Despite these limitations, the system demonstrates the viability of hybrid retrieval approaches for Indonesian news and provides a practical framework for structured information extraction without relying heavily on generative models.

4. Conclusion

This paper presents ChromoNews, a hybrid information retrieval system for Indonesian news articles. By fusing BM25 and semantic search using Reciprocal Rank Fusion, the system achieves superior performance over individual methods, with an MRR of 0.70 compared to 0.62 for BM25 and 0.65 for semantic search alone. The key innovation is the decoupling of fact extraction from narrative generation, where rule-based and NER-powered 5W1H extraction ensures factual accuracy while the LLM is used only for paraphrasing extracted facts, reducing hallucination risks and computational costs.

For future research will focus on expanding the document corpus beyond 2023 to include more recent articles and additional sources. Improving 5W1H recall through adaptive learning techniques and incorporating more sophisticated tokenization for BM25 could further enhance system performance. Developing offline alternatives to Gemini for API-independent summarization would increase system reliability. Additionally, integrating user feedback mechanisms could enable continuous improvement of the retrieval and extraction components.

References

[1] M. L. Khodra, "Ekstraksi Event 5W1H dari Teks Berita Berbahasa Indonesia," *Jurnal Cybermatika*, vol. 6, no. 1, pp. 1-12, 2018.
[2] William and Y. Sari, "CLICK-ID: A novel dataset for Indonesian clickbait headlines," *Data in Brief*, vol. 32, p. 106231, 2020.
[3] L. A. Hagen, "News mining for Homicide characteristics in Indonesia," Bachelor Thesis, Leiden Institute of Advanced Computer Science, 2024.
[4] S. Robertson and H. Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333-389, 2009.
[5] T. A. P. Aulia, "Automatic Text Summarization (ATS) pada Berita Bahasa Indonesia Berdasarkan 5W1H dengan Metode Peringkasan Ekstraktif dan Maximal Marginal Relevance (MMR)," Skripsi, Universitas Sumatera Utara, 2023.
[6] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. 2019 Conf. Empirical Methods in Natural Language Processing*, Hong Kong, 2019, pp. 3982-3992.

[7] J. A. Wallace, "Hybrid Search Benefits for RAG Systems," *Redis Blog*, Apr. 2024. [Online]. Available: https://redis.io/blog/hybrid-search-benefits-rag-systems.md. [Accessed: Nov. 15, 2024].
[8] D. Siahaan, N. P. Sutramiani, N. Suciati, I. N. Duija, and I. W. A. S. Darma, "DeepLontar dataset for handwritten Balinese character detection and syllable recognition on Lontar manuscript," *Sci Data*, vol. 9, no. 1, Dec. 2022, doi: 10.1038/s41597-022-01867-5.
[9] S. H. Khan, A. Sohail, A. Khan, and Y. S. Lee, "COVID-19 Detection in Chest X-ray Images Using a New Channel Boosted CNN," *Diagnostics*, vol. 12, no. 2, Feb. 2022, doi: 10.3390/diagnostics12020267.
[10] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, UK: Cambridge University Press, 2008.
[11] P. Sharma and R. S. Anand, "A comprehensive evaluation of deep models and optimizers for Indian sign language recognition," *Graphics and Visual Computing*, vol. 5, p. 200032, Dec. 2021, doi: 10.1016/j.gvc.2021.200032.
[12] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal Rank Fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Res. Dev. Inf. Retr. (SIGIR '09)*, Boston, MA, 2009, pp. 758-759.
[13] S. Farhadpour, T. A. Warner, and A. E. Maxwell, "Selecting and Interpreting Multiclass Loss and Accuracy Assessment Metrics for Classifications with Class Imbalance: Guidance and Best Practices," *Remote Sens (Basel)*, vol. 16, no. 3, Feb. 2024, doi: 10.3390/rs16030533.
[14] L. Alzubaidi et al., "Review of deep learning: concepts, CNN architectures, challenges, applications, future directions," *J Big Data*, vol. 8, no. 1, Dec. 2021, doi: 10.1186/s40537-021-00444-8.
[15] "Submissions Preparation Checklist," *Lontar Komputer*. [Online]. Available: https://ejournal3.unud.ac.id/index.php/lontar/about/submissions. [Accessed: Nov. 15, 2024].
[16] S. Park, J. Kim, S. Wang, and J. Kim, "Effectiveness of Image Augmentation Techniques on Non-Protective Personal Equipment Detection Using YOLOv8," *Applied Sciences (Switzerland)*, vol. 15, no. 5, Mar. 2025, doi: 10.3390/app15052631.

p-ISSN: XXXX-YYYY
e-ISSN: XXXX-YYYY